

Agenda

- Last time
 - Some distributions used over and over again
 - Bernoulli, Binomial, Geometric
- Today
 - Hypergeometric
 - Continuous distribution

First principles

$$E[X] = p$$

$$\text{Var}[X] = p(1-p)$$

$$\text{Var}[X] = \sum X_i$$

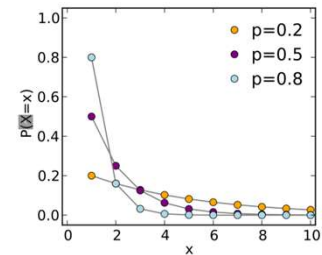
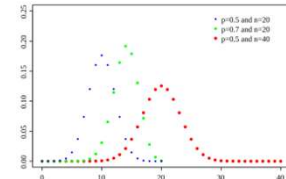
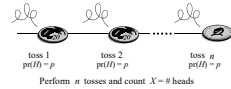
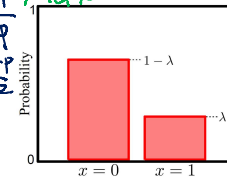
$$E[X] = np$$

$$\text{Var}[X] = np(1-p)$$

$$E[X] = 1 \text{ MGF}$$

$$\text{Var}[X] = \frac{p}{p^2}$$

toss 1
 $\text{pr}(H) = p$



2

Hypergeometric Distribution

- If we sample Y balls at random from a finite population and
 - Count X , the number of black balls
- $X \sim \text{Hypergeometric}(M, N-M, Y)$ (black, white, Y)

$$\text{Range} = \{\max(0, Y - (N - M)), \{\max(0, Y - (N - M)) + 1, \dots, \min(Y, M)\}\}$$

$$P_X(x) = \begin{cases} \frac{\binom{M}{x} \binom{N-M}{Y-x}}{\binom{N}{Y}} & \text{for } x \in \text{Range}(X) \\ 0 & \text{otherwise} \end{cases}$$

for $x \in \text{Range}(X)$
otherwise

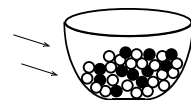
$N=10, M=7, Y=5$, then
 $\max(0, 5-3)=2$

Cannot make
all Y white

Cannot get
more black
balls than in
the bowl

Octave: `hygepdf(x, N, M, Y)`

N balls in an urn, of which there are
 M black balls
 $N-M$ white balls
Choose Y balls



3

Hypergeometric Alternate notation

- N black balls, M white balls, N+M total balls
- n items chosen at random

$$P(X = i) = \frac{\binom{N}{i} \binom{M}{n-i}}{\binom{N+M}{n}}, \quad 0 \leq i \leq \min(N, n)$$

54

4

Expectation

- Consider random variable X_i which has value 1 if the i th trial produces a black ball and 0 otherwise
- We want to compute $E[\sum X_i]$

$$\begin{aligned} P(X_1 = 1) &= \frac{N}{N+M} \\ P(X_2 = 1) &= P(X_2 = 1 \mid X_1 = 1)P(X_1 = 1) + \\ &\quad P(X_2 = 1 \mid X_1 = 0)P(X_1 = 0) \\ &= \frac{N-1}{N+M-1} \frac{N}{N+M} + \frac{N}{N+M-1} \frac{M}{N+M} = \frac{N}{N+M} \\ \text{In general, } P(X_i = 1) &= \frac{N}{N+M} \end{aligned}$$

55

5

Expectation

- Consider random variable X_i which has value 1 if the i th trial produces a black ball and 0 otherwise
- We want to compute $E[\Sigma X_i]$

$$\begin{aligned}
 P(X_1 = 1) &= \frac{N}{N+M} \\
 P(X_2 = 1) &= P(X_2 = 1 | X_1 = 1)P(X_1 = 1) + \\
 &\quad P(X_2 = 1 | X_1 = 0)P(X_1 = 0) \\
 &= \frac{N-1}{N+M-1} \frac{N}{N+M} + \frac{N}{N+M-1} \frac{M}{N+M} = \frac{N}{N+M} \\
 \text{In general, } P(X_i = 1) &= \frac{N}{N+M}
 \end{aligned}$$

55

6

Expectation

- Each X_i is a Bernoulli random variable with parameter $p = \frac{N}{N+M}$

$$\begin{aligned}
 X &= \sum_{i=1}^n X_i \\
 E(X) &= E\left(\sum_{i=1}^n X_i\right) = \frac{nN}{N+M}
 \end{aligned}$$

56

8

Variance

- Variance of each individual event can be computed easily, since Each X_i is a Bernoulli random variable with parameter $p = \frac{N}{N+M}$

$$\text{Var}(X_i) = P(X_i = 1)(1 - P(X_i = 1)) = \frac{NM}{(N + M)^2}$$

- But are these variables independent?
 - For example, X_1 and X_2 are not independent
- So we rely on

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

56

9

Covariance

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$$

$$E(X_i X_j) = P(X_i = 1, X_j = 1) = \frac{N-1}{N+M-1} \frac{N}{N+M}$$

$$\text{Cov}(X_i, X_j) = \frac{N(N-1)}{(N+M)(N+M-1)} - \left(\frac{N}{N+M}\right)^2 = \frac{-NM}{(N+M)^2(N+M-1)}$$

57

11

Variance

- X_i and X_j are not independent

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

$$\text{Var}(X) = \frac{nNM}{(N+M)^2} \left(1 - \frac{n-1}{N+M-1}\right)$$

$$= np(1-p) \left(1 - \frac{n-1}{N+M-1}\right)$$

$$\text{Var}(X) \approx np(1-p) \quad \text{when } N \text{ or } M \text{ is large}$$

56

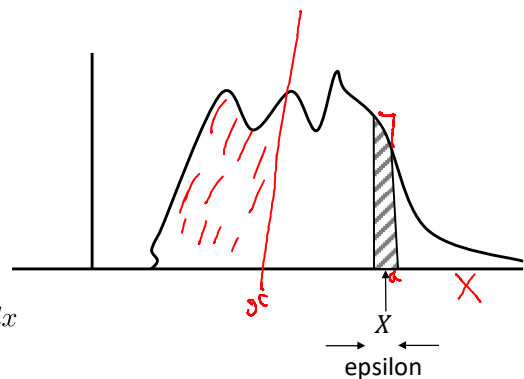
12

Continuous Distributions

- The probability density function (pdf) is the derivative of the cumulative distribution function (cdf) at that value x

$$f_X(a) = \lim_{\epsilon \rightarrow 0} \frac{\Pr\left\{a - \frac{\epsilon}{2} \leq X \leq a + \frac{\epsilon}{2}\right\}}{\epsilon}$$

$$\Pr\left\{a - \frac{\epsilon}{2} \leq X \leq a + \frac{\epsilon}{2}\right\} \approx \epsilon f(a) = \int_{a-\frac{\epsilon}{2}}^{a+\frac{\epsilon}{2}} f_X(x) dx$$



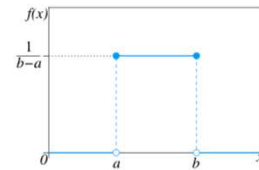
13

Uniform distribution

- A uniform random variable over the interval $[a, b]$

- valid pdf – non-negative and integrates to 1
- Mean (and median) equal to $(b + a)/2$

$$M(t) = E(e^{tX}) = \int_a^b \frac{e^{tx}}{b-a} dx = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)}, & t \neq 0 \\ 1, & t = 0 \end{cases}$$



$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

$E[X]$

$$E(X) = \int_a^b \frac{x}{b-a} dx = \frac{a+b}{2}$$

$$E(X^2) = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2 + ab + b^2}{3}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

117

15

Why uniform?

Seed... 73456

- Uniform random variables, especially over the $[0, 1]$ interval are very important, in developing programs to draw samples from other distributions
- That's the theory
 - In practice, we assume that we have access to a generator that can generate a canonical random variable
 - The random number generator is actually a pseudo-random number generated

119

16